How Task Complexity Shapes Information Search Patterns in Agents for Computer Use

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Abstract

We examined how task complexity shapes information sea rch in Agents for Computer Use (ACUs). Using Payne's inf ormation-board paradigm, we varied alternatives and infor mation dimensions in an apartment-choice task. Across 12 conditions, ACUs searched nearly all information at lower c omplexity but narrowed search sharply when both factors i ncreased. In two equal-sized board comparisons, more dim ensions were associated with greater search inequality. Th e mean search index decreased as dimensions increased but remained positive. These findings highlight the need to evaluate search processes and task structure alongside fin al outcomes.

1 Introduction

AI agents can increasingly perceive and interact directly wit h graphical user interfaces, giving them the potential to aut omate digital tasks that have traditionally required labor-int ensive human interaction [2]. However, real-world environ ments contain many interactive elements and complex task structures, making reliable performance under increasing c omplexity important to the utility of Agents for Computer Us e (ACUs).

Current ACU evaluations emphasize outcome-oriented met rics such as task success while giving less attention to task complexity in benchmark design [5]. Task complexity is an objective property of a task and should be distinguished fro m task difficulty, which emerges from interactions among th e task, performer, and context [3]. Systematically manipulat ing objective task structure makes it possible to observe ch anges in search, effort allocation, and stopping behavior dir ectly.

The information-board task developed by Payne provides a controlled method for observing these processes [4]. It inde pendently varies the numbers of alternatives and informatio n dimensions while recording each information acquisition [3, 4]. This approach enables analysis of which information i s acquired, how much information is acquired, and in what sequence. This study examined how ACUs' information-se arch patterns changed as task complexity increased.

2 Methodology

2.1 Experimental Design

This study reimplemented the information-board paradigm as a web-based apartment selection task. Participants and ACUs were instructed to "Freely explore the information ab out several apartments, then select and submit the one apa rtment you prefer most."

Alternatives were presented in rows and information dimen sions in columns. Each alternative represented a candidate apartment, and each dimension represented an attribute us ed to evaluate the apartments. All information cells were init ially masked, and their contents became visible only when clicked

	Distance from Campus	Room Size	Rent	Laundry Facilities	Natural Light	Noise Level	Included Appliances	Parking
A	5-minute walk	12 square meters	₩400,000	Nearby laundromat	Moderate	Moderate	Fully furnished	No parking
B	Click	Click	₩800,000	Click	Click	Click	Click	Click
C	Click	Click	₩800,000	Click	Click	Click	Click	Click
D	30-minute walk	17 square meters	₩400,000	Shared laundry	Low	Moderate	Basic furniture	Click
E	Click	25 square meters	₩600,000	Click	Click	Click	Click	Click
F	Click	12 square meters	₩600,000	Click	Click	Click	Click	Click
G	Click	12 square meters	₩600,000	Click	Click	Click	Click	Click
H	Click	25 square meters	₩600,000	Click	Click	Click	Click	Click

Final choice: A B C D E F G H

Figure 1. Task interface: rows = alternatives; columns = dimension s. Clicking reveals a masked cell.

The numbers of alternatives (2, 4, 8, and 12) and dimensio ns (4, 8, and 12) were fully crossed, producing 12 condition s containing 8 to 144 cells. The viewport and cell dimension s were fixed, and all task elements fit on a single screen in every condition. Because the task had no objectively correc t answer, final choices were not analyzed.

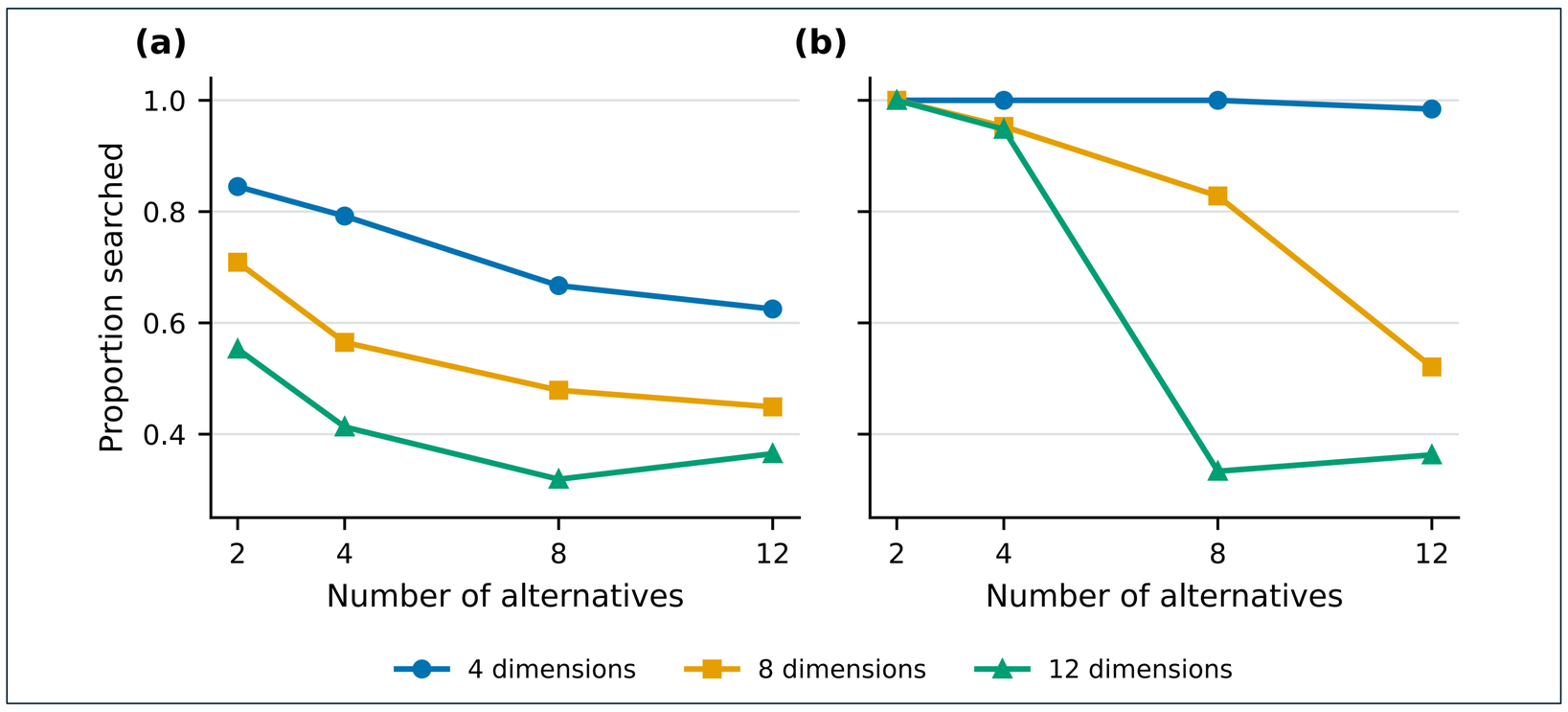


Figure 2. Proportion of information searched by (a) human participants and (b) ACUs as a function of the numbers of alternatives and dime nsions. ACU values are averages across three models.

2.2 ACU Configuration

The ACU was implemented using a tool capable of visually perceiving the GUI and directly operating the computer [1]. Models at different scales (Claude Haiku 4.5, Sonnet 4.6, a nd Opus 4.8) were selected for comparison. Through the to ol, each model could perform only one action per turn. Bec ause the task required neither page transitions nor separat e mouse movement, only click actions were used. Each mo del determined its own stopping point.

2.3 Measures

This study measured search proportion, search inequality, search direction, and API cost. Search proportion was the number of uniquely viewed cells divided by all available cell s. Search inequality represented how unevenly information viewing was distributed across alternatives and was measu red using the mean absolute deviation (MAD): MAD = (1/N) Σ $|x_i - \bar{x}|$ where x_i is the proportion of cells viewed for altern ative i, $\bar{x}$ is the mean proportion across alternatives, and N i s the number of alternatives in the session.

Transitions between consecutively viewed cells were classi fied as attribute-wise, alternative-wise, or other. Search dire ction was measured using the Search Index (SI) proposed i n prior work [4]: SI = (n_alt − n_att) / (n_alt + n_att), excludi ng other transitions. Positive and negative SI values indicat e alternative-wise and attribute-wise search, respectively. A PI cost was calculated based on token usage and model-s pecific pricing.

3 Results

3.1 Proportion Searched by Task Complexity

The ACU results were descriptively compared with the hum an proportions reported by Payne [4]. In 4-dimension condit ions, the mean proportion searched declined from .845 to . 625 as the number of alternatives increased from 2 to 12. T he corresponding ACU proportion remained near exhaustiv e, declining only from 1.000 to .984.

In the 8-dimension conditions, proportion declined from .70 9 to .449, whereas the ACU proportion declined from 1.000 to .521. In the 12-dimension conditions, the corresponding proportions declined from .553 to .365 for participants and f rom 1.000 to .363 for the ACUs. The ACU decline was parti cularly sharp between 4 and 8 alternatives in the 12-dimen sion condition. Overall, the human results showed a gradua l reduction, whereas the observed ACU runs remained nea r exhaustive at lower complexity and narrowed sharply on l arger information boards.

3.2 Search Inequality and Search Direction

Search inequality and direction were compared across the 12 conditions using values averaged across the three ACU models. With the same total number of cells, inequality was higher in conditions with more dimensions: 4 alternatives × 8 dimensions (mean MAD = .070) versus 8 × 4 (mean MA D = .000), and 4 × 12 (mean MAD = .078) versus 12 × 4 (m ean MAD = .029).

Mean SI decreased from 1.000 at 4 dimensions to .833 at 8 dimensions and .339 at 12 dimensions, indicating weaker a lternative-wise search as dimensionality increased. Informa tion-search patterns also varied across models: mean sear ch proportion ranged from .707 to .892, mean MAD from .0 22 to .119, mean SI from .590 to .870, and mean per-run A PI cost from $0.253 to $2.908.

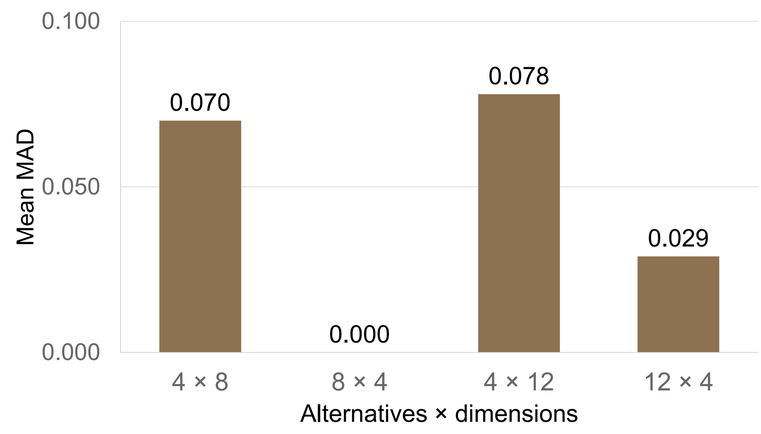


Figure 3. Equal-sized board comparisons reported in §3.2: 32 cells (first pair) and 48 cells (second pair).

4 Conclusion

This study independently manipulated the numbers of alter natives and information dimensions to observe how task co mplexity affected ACU information search. ACUs searched nearly all information at lower complexity but substantially n arrowed their search when both factors increased. In contr ast, Payne's historical human data showed a more gradual reduction in search proportion. This comparison is descripti ve.

In the two equal-sized board comparisons, conditions with more dimensions exhibited greater search inequality. This suggests that ACU search may depend on how information is structured across alternatives and dimensions, as well as on its total amount.

Evaluations should therefore consider both the size and str uctural characteristics of the information space, and exami ne search behavior alongside final task outcomes. Future r esearch should test the scope and causes of these pattern s across a variety of tasks and models.

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